## PERSONAL INFORMATION




# Dandu Purnachandra Madhukar

Berlin, Germany (Living in Berlin since several months)
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Status MSc Data Science Student at GISMA University Berlin • Resident in Berlin
Work Authorization Legal right to work in Germany (Werkstudent status • No visa sponsorship required)

## JOB APPLIED FOR

**Werkstudent:in Business Intelligence ETCS-Diagnose • Deutsche Bahn AG (Berlin 10789, Germany)**

## POSITION PERSONAL STATEMENT

Data Science Master's student at GISMA University Berlin with practical experience in machine learning pipelines, predictive modeling, and SQL database optimization. Background includes software & data engineering internship at Crack-ED (reducing query latency by 60%), data science internship at Innomatics (87% NLP intent classification), and graduate AI research. Resident in Berlin, Germany with full Werkstudent eligibility (up to 20 hrs/week during semester, 40 hrs/week during semester breaks; no visa sponsorship required). Seeking a Werkstudent or Internship role in Data Science, Machine Learning, or Data Engineering in Berlin.

## WORK EXPERIENCE

2025 – Present

### Graduate Student Researcher & AI Systems Developer

GISMA University of Applied Sciences • Berlin, Germany

- Developing predictive data science pipelines, statistical classification models, and autonomous AI agent workflows using Python, FastAPI, and Pydantic as part of M.Sc. Data Science curriculum.
- Architecting data-driven web applications and client management platforms (Auslandly CRM) utilizing PostgreSQL, Supabase, and real-time data synchronization.
- Engineering an AI-assisted Germany administrative navigation assistant integrating LLM APIs and mobile interfaces to streamline public bureaucratic workflows.

Business or sector Applied Data Science, Machine Learning & AI Systems

Sep 2024 – Jun 2025

### Software & Data Engineering Intern

Crack-ED • Gurgaon, India

- Optimised 15+ critical SQL queries and implemented Redis caching, cutting average API response time from ~800ms to under 320ms — a 60% backend latency improvement.
- Built 3 real-time analytics dashboards tracking learning metrics across 10,000+ enrolled students, helping instructors identify at-risk learners 40% faster.
- Automated CI/CD deployment workflows using GitHub Actions, cutting build-and-test deployment time from ~50 minutes to under 30 minutes (saving ~8 engineering hours weekly).

Business or sector Educational Technology & Data Analytics Platforms

Mar 2023 – Jun 2023

### Data Science & ML Intern

Innomatics Research Labs • Hyderabad, India

- Built an NLP healthcare chatbot handling 500+ query types, achieving 87% intent classification accuracy using Python and NLTK.
- Automated text preprocessing pipeline across 3 datasets totalling 50,000+ records, reducing manual data preparation time by 20% (saving ~4 hours per sprint).
- Delivered tokenization, stemming, and intent classification pipelines, reducing end-to-end chatbot inference latency by 30%.

Business or sector Artificial Intelligence & Machine Learning Research

## EDUCATION AND TRAINING

2025 – Present (Expected 2027)

### Master of Science (M.Sc.) in Data Science

EQF level 7

GISMA University of Applied Sciences • Berlin, Germany

- **Core Curriculum:** Machine Learning, Predictive Modeling, Big Data Analytics, Statistical Inference, AI Ethics.

2021 – 2024

### Bachelor of Technology (B.Tech.) in Computer Science (AI & Machine Learning)

EQF level 6

Sai Spurthi Institute of Technology • Telangana, India

- **Coursework:** Machine Learning, Deep Learning, Data Structures & Algorithms, Probability & Statistics, Database Management Systems.

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## PERSONAL SKILLS

Mother tongue(s): **Telugu** (Native)

Other language(s)

| UNDERSTANDING | | SPEAKING | | WRITING |
|---|---|---|---|---|
| Listening | Reading | Spoken interaction | Spoken production | Writing |
| **English** | | | | |
| **B2** | **B2** | **B2** | **B2** | **B2** |
| *B2 – Upper Intermediate • Professional working proficiency* | | | | |
| **Hindi** | | | | |
| **B2** | **B2** | **B2** | **B2** | **B2** |
| *B2 – Upper Intermediate* | | | | |
| **German** | | | | |
| **A2** | **A2** | **A2** | **A2** | **A2** |
| *A2 – Elementary / Conversational German • Currently learning* | | | | |

Levels: A1/A2: Basic user - B1/B2: Independent user - C1/C2: Proficient user
Common European Framework of Reference for Languages

Communication skills

- Strong technical and cross-functional communication developed by coordinating multi-role LMS platforms across students, instructors, and administrators at Crack-ED.
- Collaborative engineering within agile sprints across distributed data science teams at Innomatics Research Labs.

Organisational / managerial skills

- CI/CD sprint release management: automated GitHub Actions pipelines saving ~8 engineering hours weekly.
- Data preprocessing coordination: managed data cleaning workflows across 3 datasets and 50,000+ records.

Job-related skills

- **Machine Learning & AI:** Scikit-learn, Random Forest, Logistic Regression, basic TensorFlow, Exploratory Data Analysis (EDA), feature engineering, cross-validation.
- **Natural Language Processing:** NLTK, tokenization, stemming, text preprocessing, and intent classification pipelines (87% accuracy).
- **Backend & Database Optimization:** SQL query optimization (60% latency reduction), Redis caching, Django REST APIs, JWT + RBAC security.

Digital competence

| SELF-ASSESSMENT | | | | |
|---|---|---|---|---|
| Information processing | Communication | Content creation | Safety | Problem solving |
| Independent user (B2) | **Proficient user (C1)** | Independent user (B2) | Independent user (B2) | **Proficient user (C1)** |

Levels: Basic user - Independent user - Proficient user • Digital competences - Self-assessment grid

- **ML / AI Stack:** Scikit-learn, TensorFlow (Basic), NLTK, Random Forest, Logistic Regression, NLP
- **Data & Analytics:** Python, Pandas, NumPy, SQL, PostgreSQL, EDA, Data Cleaning, Data Visualisation
- **Backend & Cloud:** Django, REST APIs, Redis, JWT Authentication, AWS EC2/S3, CI/CD (GitHub Actions), Docker (Basic)
- **Programming Languages:** Python, JavaScript, TypeScript, SQL

## ADDITIONAL INFORMATION

Projects
Certifications
References

### Tailored Key Projects (Targeting Deutsche Bahn AG):

- **Auslandly Consultancy CRM:** Built full-featured CRM platform for relocation and study consultancy with real-time data sync, multi-step document intake, and status dashboards. • **Impact:** End-to-end delivery from database architecture to production deployment. *(Tech: React, Supabase, PostgreSQL, Realtime Sync)*
- **AI Germany Navigation App:** Developed an AI navigation assistant to help international newcomers navigate complex German bureaucratic and registration processes. • **Impact:** Streamlined administrative workflows with deterministic multilingual guidance. *(Tech: React Native, LLM APIs, Python, Mobile UX)*

### Certifications:

- **Machine Learning Specialization:** Andrew Ng, Coursera
- **Python for Data Science:** Coursera
- **SQL for Data Analysis:** Coursera

### References:

- Academic and employment references available upon request.

## ANNEXES

- GISMA University of Applied Sciences admission & enrollment verification (Berlin, Germany);
- B.Tech in Computer Science (Artificial Intelligence & Machine Learning) Degree Certificate and Transcripts;
- Course certificates for Machine Learning Specialization, Python for Data Science, and SQL for Data Analysis (Coursera);
- Verified GitHub technical repository codebases: github.com/purnachand812.

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